

# Fundamentals of Software Project Management (SE4002)

## Mid-1 Exam

Total Time(Hrs): 1  
Total Marks: 28 Points  
Total Questions: 4

Date: Sep 24<sup>th</sup> 2024

Course Instructor(s)

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Roll No

Section

Student Signature

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Attempt all the questions.

**CLO #2: Critically evaluate and discuss the issues around project management and its application in the real world with course participants and learners**

Q1 : Read and comprehend the following case study and answer the questions at the end of the case study [4-Points]

Gymo Co. is one of the world's leading health and fitness chains, with a wide range of branches in GCC. Gymo plans to launch a new innovative mobile app to address their customers (gym's members). It will offer unique in-app experiences for members. This creates a level of communication with clients like never before. Gymo will track the Progress of gym members such as: tracking steps, calories, exercise, or metrics such as heart rate and weight, visually showing your members how well they are progressing is a great way to motivate and support them. This mobile app will provide easy communication with members anytime and anywhere. Whether it's a class update, the schedule has changed, reminders, exclusive offers or personalized discounts, notification will be pushed immediately to their smartphone. Moreover, members are allowed to directly book classes and courses, monitor their payment, make secure payments, and keep up to date with their membership plan through the membership platform in the proposed app. To add more value, the mobile app will be integrated with wearable devices like Jaw bone or Android Wear to help the members stay on track of their health. Also, to build brand loyalty and keep members engaged through the mobile app, Gymo will provide valuable content including blogs, nutrition tips, podcasts, and exercise videos. Joseph Lortan, Gymo' financial controller, estimated the project's budget to be around \$100,000.

You are hired as a project manager and responsible for **creating project charter** for the Gymo.Co App. Your **project charter** contains the business reasons for the project, project objectives, project scope, project requirements, project deliverables, project budget, risks associated with the project, stack holders, measuring criteria for the objectives and project success criteria.

**Answer :** See the link : <https://www.chegg.com/homework-help/questions-and-answers/case-study-gymo-co-one-world-s-leading-health-fitness-chains-wide-range-branches-gcc-gymo--q115089195>

Q2 :Read and comprehend the following case study and answer the questions at the end of the case study [4-Points]

Jim, a final-year computer science student, has been assigned the role of project manager for her capstone project team. The team is tasked with developing a University Management System (UMS) for their university. The UMS will manage students information, course registrations, grades, and attendance. It will also have features for faculty to upload assignments, submit grades, and communicate with students. Jim knows that, as the project manager, he must carefully handle all

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aspects of the project, including planning, scheduling, resource management, and risk management. His professor advised him to follow the **PMBOK (Project Management Body of Knowledge) guidelines** and focus on the key project management knowledge areas such as **scope management, time management, cost management, quality management, risk management, communication management, and stakeholder management**. Jim started by gathering the requirements from the university's administration, faculty, and students to define the project scope. He realized that controlling scope creep is crucial to ensure that the project is completed on time. He also created a detailed schedule, budget, and communication plan to manage stakeholder expectations. As issues began to arise, Jim focused on risk management and quality control, ensuring that the project stayed on track.

Based on the case study, identify the steps Jim should take under the following project management knowledge areas:

- i. **Risk Management:** What kind of risks he identified and what steps should he take to identify, assess, and mitigate risks during the project?

**Answer :** Risk Identification, Risk Assessment, Risk Mitigation Planning, Technical Risks:, Resource Risks, Scope Creep, Time Risks, Quality Risks, Monitoring and Controlling Risks

- ii. **Cost Management:** What kind of different type key components of costs he identified and what actions should he take to help him in assessing and manage the project's cost and budget effectively?

**Answer :** Direct Costs (Development and Labour Cost), Indirect Costs (Operational Cost), Cost of Quality and Overhead Costs, Cost Estimation, Budget Development, Cost Monitoring, Cost Control and Regular Reporting

Q3 : Read and comprehend the following case study and answer the questions at the end of the case study

[6-Points]

Kim Nguyen was leading a meeting to create the work breakdown structure (WBS) for her company's IT upgrade project. This project was necessary because of several high-priority, Internet-based applications the company was developing. The IT upgrade project involved creating and implementing a plan to make all employees' IT assets meet new corporate standards within nine months. These standards specified the minimum requirements for each desktop or laptop computer, including the type of processor, amount of memory, hard disk size, type of network connection, security features, and software. Kim knew that to perform the upgrades, the project team would first have to make a detailed inventory of all the current hardware, networks, and software in the entire company of 2,000 employees. Kim had worked with other stakeholders to develop a project charter and initial scope statement. The project charter included rough cost and schedule estimates for the project and signatures of key stakeholders; the initial scope statement provided a start in defining the hardware, software, and network requirements as well as other information related to the project scope. Kim called a meeting with her project team and other stakeholders to further define the scope of the project. She wanted to get everyone's ideas on what the project involved, who would do what, and how they could avoid scope creep. The company's new CEO, Walter Schmidt, was known for keeping a close eye on major projects. The company had started using a new project management information system that let everyone know the status of projects at a detailed and high level. Kim knew that a good WBS was the foundation for scope, time, and cost performance, but she had never led a team in creating one or allocating costs based on a WBS. Help Kim in the process of **constructing WBS** for his project.

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## Answer

### 1. IT Upgrade Project

- 1. Hardware Upgrade
  - 1.1 Assess Current Hardware
  - 1.2 Determine Upgrade Requirements
  - 1.3 Purchase New Hardware
  - 1.4 Install New Hardware
  - 1.5 Test New Hardware
- 2. Software Installation
  - 2.1 Assess Current Software
  - 2.2 Determine Required Software
  - 2.3 Purchase Licenses
  - 2.4 Install Software
  - 2.5 Configure Software Settings
- 3. Network Upgrade
  - 3.1 Assess Current Network
  - 3.2 Identify Network Upgrade Needs
  - 3.3 Purchase Network Equipment
  - 3.4 Install Network Upgrades
  - 3.5 Test Network Functionality
- 4. Inventory Assessment
  - 4.1 Conduct Detailed Inventory
  - 4.2 Document Current Assets
  - 4.3 Analyze Compliance with Standards
- 5. Compliance and Quality Assurance
  - 5.1 Review Compliance Standards
  - 5.2 Implement Quality Checks
  - 5.3 Prepare Compliance Report
- 6. Training and Support
  - 6.1 Develop Training Materials
  - 6.2 Schedule Training Sessions
  - 6.3 Provide Ongoing Support

***CLO #1: Explain principles of the project lifecycle and how to identify opportunities to work with learners on relevant and appropriate project scenarios to share this understanding***

Q4 : Briefly answers the following questions :

[14-Points]

- i. Why are the Project **Scope and Integration Management** activities considered to be crucial to the success of a project ? How does **Integration management** ensure the coordination of various project components to achieve a unified outcome?

**Answer:** Project Scope Management(Scope Planning,Scope Definition,Create Work Breakdown Structure (WBS),Scope Validation, Scope Control and Monitor and Control Scope) ensures that all aspects of the project are well-defined and controlled, preventing scope creep and resource wastage. Project Integration Management, on the other hand, ensures that these defined components ( Develop Project Charter,Develop Project Management Plan,Direct and Manage Project Work,Manage Project Knowledge, Monitor and Control Project Work,Perform Integrated Change Control and Close Project or Phase ) work together cohesively toward achieving the project's objectives. Both areas are

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fundamental to project success, facilitating coordination, effective communication, and adaptive management throughout the project lifecycle.

- ii. Identify the **factors** that make the **software projects** much more difficult to manage, compared to many other types of projects.

## Answer

- I. Invisibility: With software, progress is not immediately visible
- II. Complexity: Per dollar, software products contain more complexity than other engineered artefacts
- III. Conformity: Software developers have to conform to the requirements of human clients
- IV. Flexibility: Software systems are particularly subject to change

- iii. How does proper **project planning phase** influence the success of a project? [3]

**Answer:** proper project planning lays a solid foundation for project success by providing clarity, direction, and a structured approach to managing resources, risks, finance, quality and stakeholder expectations. It enables project teams to execute efficiently, respond to challenges effectively, and achieve desired outcomes within the established constraints of time and budget.

- iv. Identify the **major key aspects** in which **modern software project management** practices differ from those of **traditional software project management**. [3]

**Answer :** There is a radical change in the basic approach taken by software industry to develop softwares over the last two decades. Hardly any software is being developed from scratch any more. Software development projects are increasingly being based on either tailoring some existing product or reusing certain pre-built libraries. Two important goals of recent life cycle models are maximization of code reuse and compression of project durations. Some other goals include Facilitating client feedbacks Accommodating customer participation, Incremental delivery, Change requests from clients are encouraged, rather than circumvented.